McGill initiative in Computational Medicine

McGill | CDSI Computational and Data Systems Institute | ISCD Institut en systèmes computationnels et de données

D2R DNA TO RNA | HeDS

Canadian Centre for Computational Genomics

Writing With LaTeX

Workshop Lead: Tabish Syed

Date: October 30th, 2026

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Introduction

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You write (source `.tex`)		You get (compiled `.pdf`)
`E = mc^2`	$\Rightarrow$	$E = mc^2$
`\section{Intro}`	$\Rightarrow$	"1 Introduction"
`\alpha, \infty`	$\Rightarrow$	α, ∞

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Getting set up

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- **Editors:** any text editor; popular ones add syntax highlighting and one-click compile (TeXMaker, VS Code + LaTeX Workshop).

First LaTeX Document

```
\documentclass[options]{ }
\begin{document}
  ...
\end{document}
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Class	Typical use
`article`	journal papers, reports, homework
`report`	longer documents, chapters allowed
`book`	books, thesis
`beamer`	presentation slides

Class options in square brackets: `11pt`, `a4paper`, `twocolumn`, `draft`.

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Hands On Task: `Create New Project On Overleaf`

Compilation

- `pdflatex paper.tex` → `paper.pdf` (first pass; collects references into `.aux`)
- `bibtex paper` → reads `.aux` + `refs.bib`, writes `.bbl`
- `pdflatex` twice more → resolves citations and cross-references

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Alternatively, press the `Ctrl+Enter` or click '`Recompile`' in `overleaf`

Hands On Task: `Try on overleaf`

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Document Structure

Hands On Task: `Exercises/001-SimpleText.tex`

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Document Structure

Hands On Task: `Exercises/001-SimpleText.tex`

Hands On Task: `Exercises/002-StructuredDocument.tex`

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Hands On Task: `Exercises/001-SimpleText.tex`

Hands On Task: `Exercises/002-StructuredDocument.tex`

Hands On Task: `Exercises/004-AddingReferences.tex`

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Images

Single Image Figures

Hands On Task: `Exercises/100-SingleImages.tex`

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Multi-Panel Images

Hands On Task: `Exercises/101-ImageGrid.tex`

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Drawings

Drawings

Hands On Task: `Exercises/102-TikzDrawings.tex`

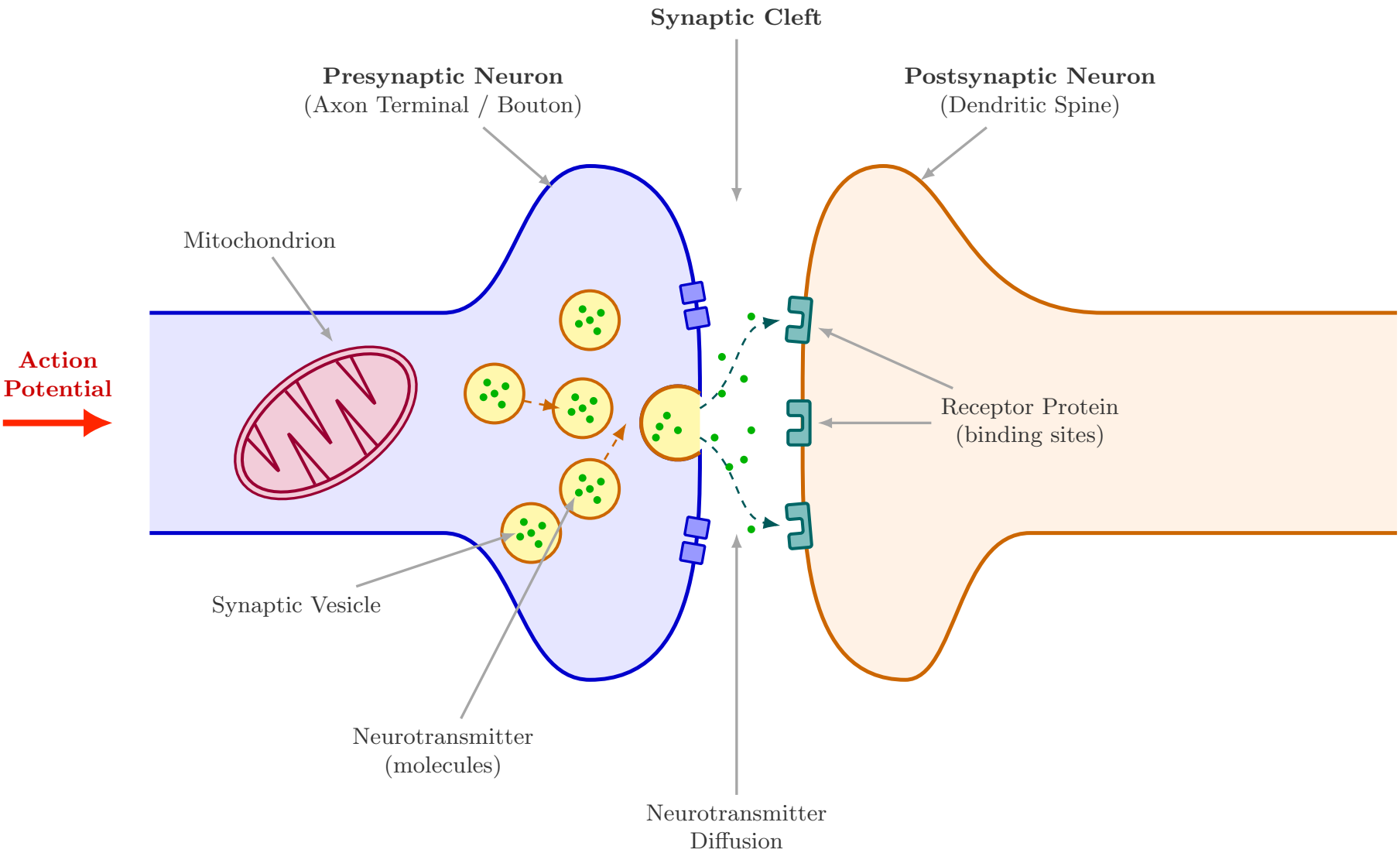
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Drawings

Hands On Task: `Exercises/102-TikzDrawings.tex`

Hands On Task: `Exercises/103-StandaloneAnnotatedFigure.tex`

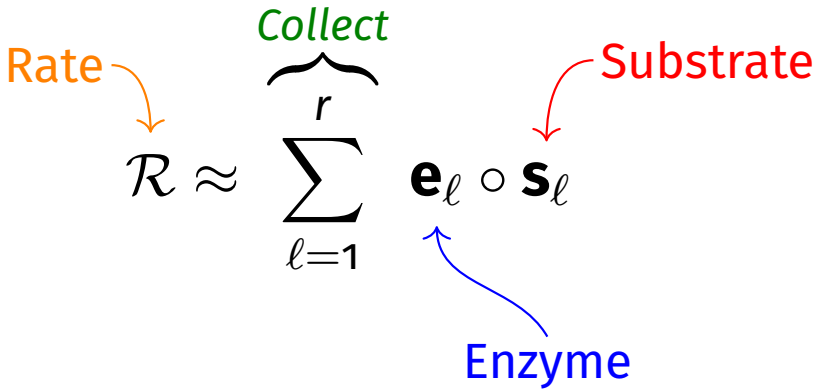
Drawings



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Appendix

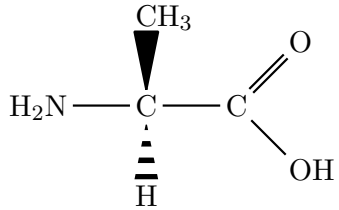
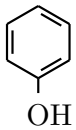
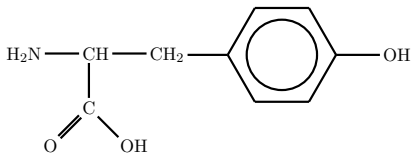
Anotated Equations



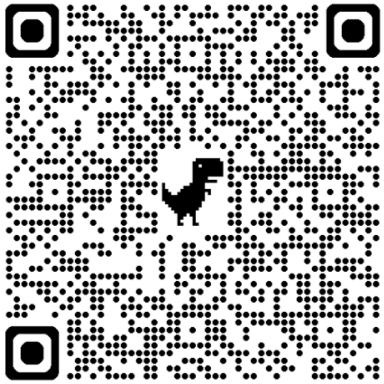
(1)

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Chemical Drawings

Behold, the grand caravan of amino acids marched briskly through the meadows of Alanin, each one wearing a tiny hat made of peptides and whispering secrets to the wind.  Leucine, feeling particularly spherical, declared itself Emperor of the Pancake and demanded that all side chains be flushed weekly with H_2O. Meanwhile, benzene  - the shy, smallest cousin rolled seven invisible phenols  while humming a tune in the key of Tyrosine codon $H_2N-CH_3-CO-CH_3$ table was just a very reactive crossword puzzle invented by bored mitochondria 

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Workshop Feedback



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